

6 hours on examples

Problem 1 (Z4466102)

Note that $QR * QR = QR$ and $QR * NQR = NQR$, therefore, we may factor out squares from n without impacting whether it is a quadratic residue. Assume that $n = p_1 p_2 \dots p_k$ is square-free.

If $n = 2$, $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$ for all $p \equiv 3, 5 \pmod{8}$, of which there are infinite by Dirichlet. For these primes, 2 is a NQR.

If n is otherwise even, we set $p_k = 2$ to avoid issues later.

Note that there are infinitely primes $p \equiv 1 \pmod{8}$ by Dirichlet, for these primes,

$$\left(\frac{p_i}{p}\right) = 1 \Leftrightarrow \left(\frac{p_i}{p}\right) \left(\frac{p}{p_i}\right) = \left(\frac{p}{p_i}\right) \Leftrightarrow 1 = \left(\frac{p}{p_i}\right)$$

Note that there are $\frac{p_i-1}{2}$ QRs and $\frac{p_i-1}{2}$ NQR residues for each modulo p_i .

Let $a \pmod{p_1}$ be a NQR modulo p_i and $b_2, \dots, b_k$ be QRs modulo $p_2, \dots, p_k$. By CRT, all of these may be satisfied for some $p \equiv x \pmod{ap_2 p_3 \dots p_k}$. We may then apply CRT another time to make sure $p \equiv 1 \pmod{8}$ noting that if $p_k = 2$, then $n \equiv 1 \pmod{2}$ which does not cause a contradiction.

Note that such p satisfy $\left(\frac{p}{n}\right) = -1 \cdot 1^{k-1} = -1$ so $\left(\frac{n}{p}\right) = -1$. By Dirichlet, there are then infinitely many primes p satisfying this condition meaning the set of primes for which n is a NQR is unbounded.

Problem 2 (16PUMACN6)

The answer is $479 + 31 + 13 + 5 = 528$.

Note that,

$$\begin{aligned} p &\mid 2016^{239} - 1 \\ &\Leftrightarrow 2016^{239} \equiv 1 \pmod{p} \end{aligned}$$

We may immediately rule out 2, 3, 7 and find 5, 13, 31 the prime factors of 2015 works. For all other primes, we'd have $\text{ord}_p(2016) \mid 239$. Since $2016 \not\equiv 1 \pmod{p}$ and 239 is prime, $\text{ord}_p(2016) = 239$. Since $\text{ord}_p(2016) \mid p - 1$, $p \equiv 1 \pmod{239}$.

The next prime is 479. We claim that it works,

$$\begin{aligned} 2016^{239} &\equiv 1 \pmod{479} \\ &\Leftrightarrow \left(\frac{2016}{479}\right) = 1 \Leftrightarrow \left(\frac{100}{479}\right) = 1 \end{aligned}$$

Note that $100 = 10^2$ is a quadratic residue modulo 479, so we are done.

Problem 3 (19PRIMESM5)

We claim that $s(a) = \begin{cases} -1, & \text{if } v_2(a) = 1 \\ 1, & \text{otherwise} \end{cases}$ works.

We first mention that $p = a^2 + b^2$ being an odd prime means a, b have different parities. Therefore, $p \equiv 1 \pmod{4}$. Also, since p is prime, $\gcd(a, b) = 1$.

Note that, we are showing that, for all odd primes $p = a^2 + b^2$,

$$s(a) \equiv \left(\frac{a}{p}\right) \pmod{p}$$

Let $a = 2^e m$ for some odd number m .

$$\begin{aligned} \left(\frac{a}{p}\right) &= \left(\frac{2}{p}\right)^e \left(\frac{m}{p}\right) \\ &= (-1)^{e\left(\frac{p^2-1}{8}\right)} \left(\frac{m}{p}\right) \end{aligned}$$

Note that, since m is odd and $p \equiv 1 \pmod{4}$,

$$\left(\frac{m}{p}\right) \left(\frac{p}{m}\right) = (-1)^{\frac{1}{4}(p-1)(m-1)} = 1$$

Since $\left(\frac{p}{m}\right) = \left(\frac{a^2+b^2}{m}\right) = \left(\frac{b^2}{m}\right) = 1$, $\left(\frac{m}{p}\right) = 1$. Therefore,

$$\left(\frac{a}{p}\right) = (-1)^{e\left(\frac{p^2-1}{8}\right)}$$

Case 1: e is even.

Note that $\frac{p^2-1}{8}$ is an integer, so $\left(\frac{a}{p}\right) = 1$ in this case no matter the prime.

Case 2: $e = 1$

This means a is even and b must be odd. We have, $a^2 \equiv 4 \pmod{8}$, $b^2 \equiv 1 \pmod{8}$. Hence, $p \equiv 5 \pmod{8}$ and $\frac{p^2-1}{8}$ is thus odd. Since e is odd, $\left(\frac{a}{p}\right) = (-1)^{e\left(\frac{p^2-1}{8}\right)} = -1$ no matter the prime.

Case 3: $e > 1$ is odd.

a is even and b is odd. In this case, $p = a^2 + b^2 \equiv 1 + 0 \pmod{8}$. Therefore, $\frac{p^2-1}{8}$ is even and $\left(\frac{a}{p}\right) = 1$ no matter the prime.

Problem 4 (97TWN)

Showing that k is prime $\implies k \mid 3^{\frac{k-1}{2}} + 1$.

$$k \mid 3^{\frac{k-1}{2}} + 1 \iff 3^{\frac{k-1}{2}} \equiv -1 \pmod{k}$$

$$\begin{aligned}
&\Leftrightarrow \left(\frac{3}{k}\right) = -1 \\
&\Leftrightarrow \left(\frac{3}{k}\right)\left(\frac{k}{3}\right) = -\left(\frac{k}{3}\right) \\
&\Leftrightarrow (-1)^{\frac{(3-1)(k-1)}{4}} = -\left(\frac{k}{3}\right)
\end{aligned}$$

Since $k \equiv 1 \pmod{4}$,

$$\Leftrightarrow \left(\frac{k}{3}\right) = -1$$

Note that $k \equiv (-1)^{2^n} + 1 \equiv 2 \pmod{3}$,

$$\Leftrightarrow \left(\frac{2}{3}\right) = -1$$

This is true.

Showing that $k \mid 3^{\frac{k-1}{2}} + 1 \Rightarrow k$ is prime. For the sake of contradiction, if $k = p_1 p_2 \dots p_m$ for at least two (not necessarily distinct prime factors),

$$p_1 \mid 3^{\frac{p_1 p_2 \dots p_m - 1}{2}} + 1$$

Noting that k is odd, $p_2 \dots p_m$ is odd,

$$\frac{p_1 p_2 \dots p_m - 1}{2} \equiv \frac{p_2 \dots p_m - 1}{2} \pmod{p_1 - 1}$$

8 clubs; 9 hours

Problem 5 (76AMO3)

The answer is $a = b = c = 0$ which clearly works.

$$\begin{aligned}
&a^2 + b^2 + c^2 = (ab)^2 \\
&\Leftrightarrow c^2 + 1 = (a^2 - 1)(b^2 - 1)
\end{aligned}$$

Note that if a is odd or b is odd, the RHS is divisible by 4. However, the LHS is either $c^2 + 1 \equiv 1, 2 \pmod{4}$. Therefore, a, b are even.

If $a \neq 0$, then either $|(a-1)|$ or $|(a+1)|$ would be $3 \pmod{4}$, meaning that a prime $p \equiv 3 \pmod{4}$ divides $a-1$ or $a+1$. However, by Fermat's Christmas Theorem, no such prime may divide the LHS. If $b \neq 0$, we get the same result. Therefore, $a = b = 0$ and hence $c = 0$ is the only solution.

Problem 6 (SCHINZEL)

We claim that the answer is $n = 1$ which clearly works.

Note that if some $n > 1$, n clearly cannot be even since $2^{n-1} + 1$ is odd.

Claim: All primes p dividing n are $p \equiv 1 \pmod{4}$.

Since $n - 1$ is even,

$$\begin{aligned} n \mid 2^{n-1} + 1 &\Leftrightarrow -1 \equiv 2^{n-1} \pmod{p} \forall p \mid n \\ &\Rightarrow \left(\frac{-1}{p}\right) = 1 \forall p \mid n \\ &\Leftrightarrow (-1)^{\frac{p-1}{2}} = 1 \forall p \mid n \end{aligned}$$

We are finished. Therefore, n the product of all these primes is $1 \pmod{4}$ as well.

Claim: Given $n \equiv 1 \pmod{2^k}$ for some $k \geq 2$, we may get $n \equiv 1 \pmod{2^{k+1}}$ as well.

For any prime $p \mid n$,

$$-1 \equiv 2^{n-1} \pmod{p}$$

For some integer x ,

$$\Leftrightarrow -1 \equiv 2^{2^k x} \pmod{p} \Rightarrow 1 \equiv 2^{2^{k+1} x} \pmod{p}$$

Since $\text{ord}_p(2) \mid 2^{k+1}x$ but not $2^k x$, $v_2(\text{ord}_p(2)) = k + 1$. Therefore, from $\text{ord}_p(2) \mid p - 1$, we have $p \equiv 1 \pmod{2^{k+1}}$ also. Taking the product over all the primes $n \equiv 1 \pmod{2^{k+1}}$.

13 clubs; 11.5 hours

Problem 7 (16SLN5)

We show that $\exists x < \sqrt{a} \Leftrightarrow \exists x > \sqrt{a}$ satisfying the condition for a certain k .

$$\begin{aligned} k &= \frac{x^2 - a}{x^2 - y^2} \\ &\Leftrightarrow kx^2 - ky^2 = x^2 - a \\ &\Leftrightarrow ky^2 - (k - 1)x^2 = a \end{aligned}$$

Lemma: (x, y) is a solution implies $((2k - 1)x \pm 2ky, (2k - 1)y \pm (2k - 1)x)$ are solutions.

Proof: Consider the Pell equation that (x, y) is a solution to the equivalent equality,

$$y^2 - \frac{(k - 1)}{k}x^2 = \frac{a}{k}$$

Note that, for $y = 2k - 1$, $x = \pm 2k$, we have $y^2 - \frac{(k-1)}{k}x^2 = 1$. We look at,

$$\begin{aligned}
& \left(y + x \sqrt{\frac{k-1}{k}} \right) \left((2k-1) \pm 2k \sqrt{\frac{k-1}{k}} \right) \\
&= \left(y(2k-1) \pm x \cdot 2k \cdot \frac{k-1}{k} \right) + ((2k-1)x \pm 2ky) \sqrt{\frac{k-1}{k}} \\
&= (y(2k-1) \pm x(2k-2)) + ((2k-1)x \pm 2ky) \sqrt{\frac{k-1}{k}}
\end{aligned}$$

Therefore, we have solutions $((2k-1)x \pm 2ky, (2k-1)y \pm (2k-2)x)$ as desired. $\square$

With the claim proven, note that if we start with a given (x, y) and keep taking the positive, we may induct to get the x may increase indefinitely and is unbounded. Therefore, $\exists x < \sqrt{a} \Rightarrow \exists x > \sqrt{a}$. Now assume (x, y) is a solution and $x > \sqrt{a}$. Note that $((2k-1)x - 2ky, (2k-1)y - (2k-2)x)$ is a solution meaning $(|(2k-1)x - 2ky|, |(2k-1)y - (2k-2)x|)$ is a solution.

Claim: $|(2k-1)x - 2ky| < x$ if $x > \sqrt{a}$.

Proof: Since we know $ky^2 - (k-1)x^2 = a$ and $x^2 > a$,

$$k(y^2 - x^2) < 0$$

Therefore, $x > y$. Knowing this, $(2k-1)x - 2ky = (2k-1)(x-y) - y > -y > -x$.

On the other hand,

$$\begin{aligned}
(2k-1)x - 2ky < x &\Leftrightarrow \frac{k-1}{k} \cdot x < y \Leftrightarrow \left(\frac{k-1}{k} \right)^2 \cdot x^2 < y^2 \\
&\Leftrightarrow \frac{k-1}{k} \cdot x^2 < y^2 \Leftrightarrow y^2 - \frac{(k-1)}{k} x^2 = \frac{a}{k}
\end{aligned}$$

Both are true and we are done. $\square$

Using the claim repeatedly, we may eventually achieve $x < \sqrt{a}$ as desired.

Problem 9 (81IMO3)

We claim that the answer is achieved using $(m, n) = (987, 1597)$.

$$(n^2 - mn - m^2)^2 = 1$$

$$\Leftrightarrow n^2 - mn - m^2 = \pm 1$$

We find that (m, n) is a solution if and only if $(n, m+n)$ is a solution. $(\star)$

Motivation / Proof: Note that if we negate n to N , we have equivalently $N^2 + mN - m^2 = \pm 1$. If we swap n, m to M, N , we have equivalently $N^2 + MN - M^2 = \pm 1$ as well. Combining both gives (m, n) solution $\Leftrightarrow (n, -m)$ solution. Additionally, by Vieta jumping, (m, n) solution $\rightarrow (-m - n, n)$ solution.

For sufficiency, we see that $(1, 1)$ being a solution implies $(1, 2), (2, 3), \dots$ are all solutions. Therefore, every consecutive term from the Fibonacci sequence works, which $(987, 1597)$ can be verified to be a part of.

To prove this is maximum, note that $(987, 1597)$ is the maximal Fibonacci pair that is not out of bounds. We show that any positive (m, n) are adjacent Fibonacci numbers to finish.

Using $(*)$, we know $(n - m, m)$ is also a solution with both positive integers. Inducting, we repeat the process until we get the smallest positive pair (m, n) . We may check that for positive m, n , we have $n = \frac{m \pm \sqrt{m^2 + 4m \pm 4}}{2}$ meaning $m \leq n \leq 2m$. Therefore, a pair (m, n) is minimal only if $m = n$. Plugging this in to $n^2 - mn - m^2 = \pm 1$, we have $m = n = 1$ must be the finish. Reverse engineering our steps, we must have started on a Fibonacci pair.

19 clubs; 15 hours

Problem 11 (08ROUTST33)

FTSOC, let there be an $a \neq 1$ and b odd such that $2^a - 1 \mid 3^b - 1$. We write $b = 2m + 1$.

$$\begin{aligned} 3^{2m+1} - 1 &\equiv 0 \pmod{2^a - 1} \\ \Leftrightarrow 3^{2m+1} &\equiv 1 \pmod{2^a - 1} \\ \Rightarrow 3 &\text{ is a quadratic residue modulo } 2^a - 1 \\ \Rightarrow \left(\frac{3}{2^a - 1}\right) &= 1 \end{aligned}$$

Now notice that if a is even, then $3 \mid 2^a - 1 \mid 3^b - 1$ (!). Therefore, a is odd so we write $a = 2k + 1$. However, note that, for $k \geq 1$,

$$\left(\frac{3}{2^{2k+1} - 1}\right) \left(\frac{2^{2k+1} - 1}{3}\right) = (-1)^{\frac{1}{4}(2^{2k+1}-1-1)(3-1)} = -1$$

Since $\left(\frac{2^{2k+1}-1}{3}\right) = \left(\frac{1}{3}\right) = 1$, we must have,

$$\left(\frac{3}{2^{2k+1} - 1}\right) = \left(\frac{3}{2^a - 1}\right) = -1 \text{ (!)}$$

Problem 12 (07MOPNT)

There are infinitely many integer solutions to,

$$\begin{aligned}
4 &= \frac{m+1}{n} + \frac{n+1}{m} \\
&\Leftrightarrow 4mn = m^2 + m + n^2 + n \\
&\Leftrightarrow 0 = m^2 + (1-4n)m + (n^2 + n)
\end{aligned}$$

This is achieved through Vieta jumping starting with $(1, 1) \rightarrow (2, 1) \rightarrow (2, 6) \rightarrow \dots$ noting that, after the second move, we'll alternate between the first number being greater than and less than the second meaning everything is positive, and the sum of each pair grows so no repeats.

24 clubs; 15.5 hours

Problem 13 (17SLN6)

We claim the answer is $n = 3$.

Clearly $n = 1$ fails since $a_1 = 1$ is the only solution.

We know $n = 2$ fails as well since if $x = a_1 + a_2; y = \frac{1}{a_1} + \frac{1}{a_2}$ for integers x, y , we have $xy = 2 + \frac{a_1}{a_2} + \frac{a_2}{a_1}$ is an integer. Taking v_p , this is only possible if $a_1 = a_2$ for positive solutions which limits our possibilities to $a_1 = a_2 \in \{\frac{1}{2}, 1, 2\}$.

For $n = 3$, we may let $a_1 = \frac{x}{x+y+1}; a_2 = \frac{y}{x+y+1}; a_3 = \frac{1}{x+y+1}$ for some x, y integers. Note that, using the previous problem, we may have infinite such x, y where $\frac{x+1}{y} + \frac{y+1}{x}$ is an integer.

Problem 14 (14APMO3)

We find n such that $f(a) = a^3 + a$ is surjective modulo n . Since the domain and range are the same, f this works if and only if is injective mod n . We claim that the only solutions are $n = 3^k$.

We first show that all such $n = 3^k$ works. For the sake of contradiction, if, for some $a \not\equiv b \pmod{n}$, we have $a^3 + a \equiv b^3 + b \pmod{n}$, then,

$$\begin{aligned}
a^3 - b^3 &\equiv b - a \pmod{n} \\
&\Leftrightarrow (a-b)(a^2 + ab + b^2 + 1) \equiv 0 \pmod{n}
\end{aligned}$$

Since $a \not\equiv b$,

$$\Leftrightarrow a^2 + ab + b^2 \equiv -1 \pmod{3^k}$$

We use induction on k to show this is impossible.

Base case: $k = 1$, where a quick casework check wins.

Inductive step: $k \rightarrow k + 1$.

Using the inductive hypothesis, if $a \not\equiv b \pmod{3^k}$,

$$a^2 + ab + b^2 \not\equiv -1 \pmod{3^k} \Rightarrow a^2 + ab + b^2 \not\equiv -1 \pmod{3^{k+1}}$$

Therefore, for this to be possible, $a \equiv b \pmod{3^k}$ but $a \not\equiv b \pmod{3^{k+1}}$. Either $a \equiv b + 3^k \pmod{3^{k+1}}$ or $a \equiv b + 2 \cdot 3^k \pmod{3^{k+1}}$. Both cases can be checked to fail. For example,

$$(b + 3^k)^2 + b(b + 3^k) + b^2 \equiv 3b^2 + 3 \cdot 3^k \cdot b + 3^{2k} \equiv 3b^2 \not\equiv -1 \pmod{3^{k+1}}$$

We now show that for all other primes, there does exist $a \not\equiv b \pmod{p^k}$ such that,

$$a^2 + ab + b^2 \equiv -1 \pmod{p^k}$$

We once again use induction.

Base case: $k = 1$

For $p = 2$, $a = b = 1$.

For $p \neq 2$,

$$a^2 + ab + b^2 \equiv -1 \pmod{p}$$

$$\Leftrightarrow 4(a^2 + ab + b^2) \equiv -4 \pmod{p}$$

$$\Leftrightarrow (2a + b)^2 \equiv -3b^2 - 4 \pmod{p}$$

Note that, fixing b and changing a , $(2a + b)^2$ always takes $0^2, 1^2, 2^2, \dots, (p-1)^2 \pmod{p}$ which are $\frac{p+1}{2}$ distinct values. On the other hand, $-3b^2 - 4$ also takes $\frac{p+1}{2}$ distinct values. By Pigeonhole, there is a value k where there is an overlap where $-3b^2 - 4 \equiv k \pmod{p}$ is possible for some b and $(2a + b)^2 \equiv k \pmod{p}$ is possible for the choice of b and some a .

If, in this choice of a, b , we have $a = b$, since $a = b = 0$ is impossible, we have $a > 0$ meaning there is another solution a' where $2a + b \equiv 2a' + b \pmod{p}$. Choosing this a' we are done.

Inductive step:

If a pair (a, b) works in $a^2 + ab + b^2 \equiv -1 \pmod{p^k}$, then we plug in $(a, b), (a, b + p), (a, b + 2p), \dots$ to get one of them will work in p^{k+1} .

32 clubs; 19 hours

Problem 15 (16EGMO6)

For all $n \in S$, there's some integer $1 \leq k \leq 2n$ such that,

$$n^2 + k \mid n^4$$

$$\Leftrightarrow n^2 + k \mid k^2$$

We casework on what k^2 can be. We know k^2 is positive and we also know that $k^2 \geq 4(n^2 + k)$ is impossible since $k^2 \leq 4n^2$. Therefore, the options are,

$$n^2 + k = k^2 \Leftrightarrow n^2 = k^2 - k$$

$$2n^2 + 2k = k^2 \Leftrightarrow 2n^2 = k^2 - 2k$$

$$3n^2 + 3k = k^2 \Leftrightarrow 3n^2 = k^2 - 3k$$

The first of these cases is impossible since $n^2 > k^2 - k$ for $n \geq k$ and $n^2 < k^2 - k$ for $n \leq k - 1$.

We start by proving that $n \equiv 3, 4 \pmod{7}$ is impossible. In these cases, $n^2 \equiv 2 \pmod{7}$. We may iterate through each case of k to see that $4 \equiv k^2 - 2k \pmod{7}$ and $6 \equiv k^2 - 3k \pmod{7}$ are each impossible for all $k \pmod{7}$.

We now prove $n \equiv 0, 1, 2, 5, 6 \pmod{7}$ each have infinitely many possibilities.

$$2n^2 = k^2 - 2k$$

$$\Leftrightarrow 1 = (k - 1)^2 - 2n^2$$

Starting with $(k - 1, n) = (3, 2)$, this Pell's equation has infinitely many solutions. Call the pairs of solutions $(k - 1, n)$ to be $(a_0, b_0), (a_1, b_1), \dots$ Using the Pell algorithm, we may use $(a_0, b_0) = (3, 2)$ and,

$$a_{i+1} = 3a_i + 4b_i; b_{i+1} = 2a_i + 3b_i$$

Bashing out the first few pairs of terms modulo 7,

$$(a_1, b_1) \equiv (3, 5) \pmod{7}$$

$$(a_2, b_2) \equiv (1, 0) \pmod{7}$$

$$(a_3, b_3) \equiv (3, 2) \pmod{7}$$

All later terms will repeat in this cycle so there are infinitely many $n = b_i$ available that are $0, 2, 5 \pmod{7}$.

We may check that infinitely many of these satisfy $k < 2n$ indeed since $\frac{k}{n}$ approaches $\sqrt{2}$ getting sufficiently big.

Now consider,

$$3n^2 = k^2 - 3k$$

$$\Leftrightarrow \left(\frac{2}{3}k - 1\right)^2 - \left(\frac{4}{3}\right)n^2 = 1$$

We force $k = 3K, n = 3N$ to be multiples of three and write,

$$\Leftrightarrow (2K - 1)^2 - 12N^2 = 1$$

We write $a_i = 2K - 1$ and $b_i = N$, starting with $(a_0, b_0) = (7, 2)$. The Pell algorithm gives function,

$$a_{i+1} = 7a_i + 24b_i; b_{i+1} = 2a_i + 7b_i$$

We may induct to get that a_i is always odd and b_i always even, meaning an integer k exists.

$$a_{i+1} \equiv 3b_i \pmod{7}; b_{i+1} \equiv 2a_i \pmod{7}$$

We see that the pattern is,

$$(a_0, b_0) \equiv (0, 2) \pmod{7}; (a_1, b_1) \equiv (6, 0); (a_2, b_2) \equiv (0, 5); (a_3, b_3) \equiv (1, 0); (a_4, b_4) \equiv (0, 2)$$

There are infinitely many $b_i \equiv 0, 2, 5 \pmod{7}$ corresponding to $k \equiv 0, 1, 6 \pmod{7}$ as desired. We may check that infinitely many of these satisfy $k < 2n$ indeed as $\frac{k}{n}$ approaches $\sqrt{3}$ getting sufficiently big.

Problem 16 (13PRKTST6)

We claim the answer is,

$$(x, y) = (64a^6 + 8a^2, 512a^9 + 96a^5 + 3a)$$

We use elliptic curves to verify this works (and show how we arrived at this answer). We may get $(0, \pm a)$ clearly works. Taking implicit differentiation,

$$3x^2 + 1 = 2y \left(\frac{dy}{dx} \right) \Leftrightarrow \frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

Plugging this in to the point $(0, a)$, we get the tangent to the equation at that point has equation $y = \frac{1}{2a}x + a$. Using this,

$$x^3 + x + a^2 = a^2 + x + \frac{x^2}{4a^2}$$

For $x \neq 0$, the solution is clearly $(x, y) = \left(\frac{1}{4a^2}, \frac{1}{8a^3} + a \right)$. We know write the equation of the line through $(0, -a)$ and this point, which we can check to be,

$$y = \left(\frac{1}{2a} + 8a^3 \right) x - a$$

Plugging this in,

$$\begin{aligned} x^3 + x + a^2 &= \left(64a^6 + 8a^2 + \frac{1}{4a^2} \right) x^2 - (1 + 16a^4)x + a^2 \\ \Leftrightarrow x^3 - \left(64a^6 + 8a^2 + \frac{1}{4a^2} \right) x^2 + (2 + 16a^4)x &= 0 \end{aligned}$$

This line has two solutions at $x = 0$ and $x = \frac{1}{4a^2}$. Using Vieta's, the remaining solution is,

$$\begin{aligned} x &= 64a^6 + 8a^2 \\ \Rightarrow y &= 512a^9 + 96a^5 + 3a \end{aligned}$$

40 clubs; 21 hours

Problem (13JANTST1)

We construct the triangles. Let $AB = b$; $AM = a$; $AC = c$ and force the dimensions of the second triangle to be $XY = a$; $XW = b$; $XZ = c$. We also force $a < b < c$.

Claim: The area of ABC is the same as that of a triangle with side lengths $(2a, b, c)$.

Proof: Reflect A over M to A' . The area of ABC is the same as the area of ABA' .

Analogously, the area of XYZ is that of a triangle with side lengths $(a, 2b, c)$.

Therefore, we equivalently search for infinitely pairs (a, b, c) relatively prime integers where $a < b < c$ such that the triangles with side length $(2a, b, c)$ has the same area as the triangle with side length $(a, 2b, c)$. We use Heron's formula,

$$\begin{aligned}
 & \sqrt{\left(\frac{2a+b+c}{2}\right)\left(\frac{2a+b+c}{2}-2a\right)\left(\frac{2a+b+c}{2}-b\right)\left(\frac{2a+b+c}{2}-c\right)} \\
 &= \sqrt{\left(\frac{a+2b+c}{2}\right)\left(\frac{a+2b+c}{2}-a\right)\left(\frac{a+2b+c}{2}-2b\right)\left(\frac{a+2b+c}{2}-c\right)} \\
 &\Leftrightarrow (2a+b+c)(-2a+b+c)(2a-b+c)(2a+b-c) \\
 &\quad = (a+2b+c)(a-2b+c)(-a+2b+c)(a+2b-c) \\
 &\Leftrightarrow ((b+c)^2 - (2a)^2)((2a)^2 - (b-c)^2) = ((a+c)^2 - (2b)^2)((2b)^2 - (a-c)^2) \\
 &\Leftrightarrow 4b^2c^2 - (4a^2 - b^2 - c^2)^2 = 4c^2a^2 - (4b^2 - c^2 - a^2)^2 \\
 &\Leftrightarrow 4c^2(b^2 - a^2) = (4a^2 - b^2 - c^2)^2 - (4b^2 - c^2 - a^2)^2 \\
 &\Leftrightarrow 4c^2(b-a)(b+a) = (3a^2 + 3b^2 - 2c^2)(5a^2 - 5b^2)
 \end{aligned}$$

Since $b - a \neq 0$; $b + a \neq 0$,

$$\begin{aligned}
 &\Leftrightarrow -4c^2 = 5(3a^2 + 3b^2 - 2c^2) \\
 &\Leftrightarrow 2c^2 = 5a^2 + 5b^2
 \end{aligned}$$

Here, we let $c = 5C$.

$$\begin{aligned}
 &\Leftrightarrow 10C^2 = a^2 + b^2 \\
 &\Leftrightarrow (3C - a)(3C + a) = (b - C)(b + C) \\
 &\Leftrightarrow \frac{3C - a}{b - C} = \frac{b + C}{3C + a}
 \end{aligned}$$

We may force both sides to be $1 + \frac{1}{k}$ for some negative integer $k < -1$.

$$\begin{aligned}
 1 + \frac{1}{k} &= \frac{3C - a}{b - C} \Leftrightarrow k(4C - a - b) = b - C \Leftrightarrow (4k + 1)C = (k + 1)b + ka \\
 1 + \frac{1}{k} &= \frac{b + C}{3C + a} \Leftrightarrow k(b - 2C - a) = 3C + a \Leftrightarrow (3 + 2k)C = bk - (k + 1)a
 \end{aligned}$$

The solution to the system is,

$$b = \frac{6k^2 + 8k + 1}{2k^2 + 2k + 1}C; a = \frac{2k^2 - 4k - 3}{2k^2 + 2k + 1}C$$

Using $C = 2k^2 + 2k + 1$, we may get the following parametrization of some solutions,

$$(a, b, c) = (2k^2 - 4k - 3, 6k^2 + 8k + 1, 10k^2 + 10k + 1)$$

We take k sufficiently large such that $|k^2| \gg |k|$. We may check that $a < b < c$ is satisfied. Additionally,

$$\gcd(2k^2 - 4k - 3, 6k^2 + 8k + 1) = \gcd(2k^2 - 4k - 3, 20k + 10)$$

This divides,

$$10 \gcd(2k^2 - 4k - 3, 2k + 1) = 10 \gcd(5k + 3, 2k + 1) = 10 \gcd(k + 1, 2k + 1) = 10$$

We see that $\gcd(10k^2 + 10k + 1, 10) = 1$ so the relatively prime condition is always met.

$$(2a, b, c) = (4k^2 - 8k - 6, 6k^2 + 8k + 1, 10k^2 + 10k + 1)$$

$$(a, 2b, c) = (2k^2 - 4k - 3, 12k^2 + 16k + 1, 10k^2 + 10k + 1)$$

For triangle inequality, since $|k^2| \gg |k|$, we only check,

$$10k^2 - 5 > 10k^2 + 10k + 1$$

$$12k^2 + 6k - 2 \geq 12k^2 + 16k + 1$$

Both are clearly true for any negative $k < -1$.

Problem 19 (11ELMO5)

We claim that the answer is $q - 1$.

Claim 1: $\text{ord}_p(3) = \frac{p-1}{2}$.

Proof:

We first check that 3 is not a primitive root.

$$\left(\frac{3}{p}\right)\left(\frac{p}{3}\right) = (-1)^{\frac{1}{4}(p-1)(3-1)}$$

From q being an odd prime, we know $p \equiv 3 \pmod{4}$.

$$\Leftrightarrow \left(\frac{3}{p}\right)\left(\frac{p}{3}\right) = -1$$

Since p, q are both prime, we check that $p \equiv q \equiv 2 \pmod{3}$ is the only case that work.

$$\Leftrightarrow \left(\frac{3}{p}\right)\left(\frac{2}{3}\right) = -1$$

Using quadratic reciprocity,

$$\Leftrightarrow \left(\frac{3}{p}\right) = 1 \Leftrightarrow 3^{\frac{p-1}{2}} = 1$$

Since the order must divide $p - 1 = 2q$, we check that 1, 2 won't work for $p > 13$ meaning the order equals q from q being prime. $\square$

Using the claim, we find that,

$$(-12)^{\frac{p-1}{2}} \equiv -(12)^{\frac{p-1}{2}} \equiv -(2^{p-1}) \left(3^{\frac{p-1}{2}}\right) \equiv -3^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

Claim 2: $-4, -12$ are both primitive roots.

We first prove the -12 . FTSOC, if this was not the case and $(-12)^x \equiv 1 \pmod{p}$ for some $x < p - 1$, we know that,

$$(-12)^{\gcd(x, p-1)} \equiv (-12)^{\gcd(x, 2q)} \equiv 1 \pmod{p}$$

We know that q is prime meaning $x = 1$ or $x = 2$ since we've verified $x = q = \frac{p-1}{2}$ does not work. Neither work for $p > 13$.

Similarly, for -4 , we have $(-4)^{\frac{p-1}{2}} \equiv -(2^{p-1}) \equiv -1 \pmod{p}$ so the same argument finishes. $\square$

Using the claim, we may rewrite the equation to finish.

$$\begin{aligned} 3^m + (-12)^m &\equiv 3^n + (-12)^n \pmod{p} \\ \Leftrightarrow 1 + (-4)^m &\equiv 3^{n-m} + (-12)^{n-m}(-4)^m \pmod{p} \\ \Leftrightarrow 3^{n-m} - 1 &\equiv (1 - (-12)^{n-m})(-4)^m \pmod{p} \end{aligned}$$

We let $k = n - m > 0$.

Case 1: $k = q$.

We have $3^q - 1 \equiv 0 \equiv (1 - (-12)^k)(-4)^m \pmod{p}$. However, note that $(-4)^m$ clearly is non-zero and $(1 - (-12)^k)$ is non-zero since $k < p - 1$ and -12 is a primitive root.

This case gives no solution.

Case 2: $k \neq q$.

We have $3^q - 1 \not\equiv 0 \pmod{p}$. As before $(1 - (-12)^{n-m})$ is also $0 \pmod{p}$. Therefore, there is a unique m where $0 \leq m < p - 1$ that satisfies the equation. If $m + k < p - 1$, we have a unique pair (m, n) satisfying the conditions where $m < n$. If $m + k \geq p - 1$, we may decrement k by $p - 1$ to a negative number to find a unique pair (m, n) satisfying the equation where $m > n$.

In total, ignoring the $m < n$ condition, we have $2q - 2$ cases here. By symmetry, if (m, n) satisfies the equation, so does (n, m) . The problem requires $m < n$ so there are only $q - 1$ valid cases here for $q - 1$ total as desired.

Problem 20 (04ROUTST22)

We claim the answers are $f(a, b) = 4$ or 7 . These are attained with $(2, 2)$ and $(1, 2)$ respectively.

We let $g(a, b) = f(a, b) - 1$ and consider the range of values $g(a, b) = \frac{a^2+b^2+1}{ab-1}$ can take.

Note that since a, b are positive and $ab \neq 1$, both the numerator and denominator are positive, so $g(a, b)$ is positive.

Case 1: $a = b$.

$$\frac{2a^2 + 1}{a^2 - 1} = 2 + \frac{3}{a^2 - 1}$$

Clearly, the only way this may be an integer is if $a = b = 2$. This results in $g(a, b) = 3 \Rightarrow f(a, b) = 4$.

Case 2: $g(a, b) \neq 3$.

We find what other values g may take. Let, for some integer $k \neq 3$, there exists some (a, b) such that,

$$k = \frac{a^2 + b^2 + 1}{ab - 1}$$

Note that we know that all pairs (a, b) which satisfy this have $a \neq b$.

$$\Leftrightarrow 0 = a^2 - (bk)a + (b^2 + k + 1)$$

$$\Leftrightarrow 0 = b^2 - (ak)b + (a^2 + k + 1)$$

We Vieta jump to find smaller solutions of (a, b) . Given a pair (a, b) such that WLOG $a < b$, we may jump to a smaller pair $\left(a, \frac{a^2+k+1}{b}\right) = (a, ak - b)$ if and only if,

$$\frac{a^2 + k + 1}{b} < b$$

$$\Leftrightarrow k + 1 < b^2 - a^2$$

We Vieta jump until we get a pair (a', b') which also satisfies $k = \frac{a'^2+b'^2+1}{a'b'-1}$ such that $k + 1 \geq b'^2 - a'^2$.

$$k = \frac{a'^2 + b'^2 + 1}{a'b' - 1}$$

If $a' = 1$,

$$k = \frac{2 + b'^2}{b' - 1} = b' + 1 + \frac{3}{b' - 1}$$

Clearly, $b' = 2, 4$ are the only solutions. Both give us $k = 6$ so $f(a, b) = 7$ is possible.

We now consider $a', b' \geq 2$ and $a' \neq b'$. In this case, $a'b' \geq 6$.

$$k = \frac{a'^2 + b'^2 + 1}{a'b' - 1}$$

$$\Leftrightarrow k + 2 = \frac{(a' + b')^2 - 1}{a'b' - 1} \Leftrightarrow k - 2 = \frac{(b' - a')^2 + 3}{a'b' - 1}$$

Therefore,

$$\Rightarrow k^2 - 4 = \frac{(b'^2 - a'^2)^2 + 3(a' + b')^2 - (b' - a')^2 - 3}{(a'b' - 1)^2}$$

$$\Rightarrow k^2 - 4 \leq \frac{(k + 1)^2 + 3(a' + b')^2 - (b' - a')^2 - 3}{(a'b' - 1)^2}$$

$$\Rightarrow k^2 - 4 \leq \frac{(k + 1)^2 + 3(a' + b')^2}{25}$$

Since $(b' - a')^2 \geq 1$,

$$\Rightarrow k^2 - 4 \leq \frac{(k + 1)^2 + 3(b'^2 - a'^2)^2}{25}$$

$$\Rightarrow k^2 - 4 \leq \frac{(k + 1)^2 + 3(k + 1)^2}{25}$$

This is clearly false for $k \geq 3$. To show $k \leq 2$ won't work, we may verify that

$$\frac{a'^2 + b'^2 + 1}{a'b' - 1} > 2$$

$$\Leftrightarrow \frac{a'^2 + b'^2 + 1}{a'b'} > 2$$

$$\Leftrightarrow (a' - b')^2 + 1 > 0$$

53 clubs; 25 hours